

U.S. Utility Patent Application (Non-Provisional) Draft

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Title: Geometric Control Conductor for Composite Materials – TitanA16

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Cross-Reference to Related Applications:

This application claims priority to U.S. Provisional Patent Application No. 63/955,998, filed January 7, 2026.

Field of the Invention

The present invention relates to electrical conductors and composite materials. More specifically, it concerns the use of geometric constraint to enhance electron flow in disordered conductive materials, applicable to wires, batteries, and other systems requiring controlled conductivity.

Background of the Invention

In 1890s, Lodge demonstrated the Coherer Effect, wherein chaotic metal filings in a tube exhibit a sharp drop in resistance when aligned by electromagnetic influence. Recent discoveries (Max Planck Institute, 2025) show that quantum geometry alone can separate chiral fermions without magnetic fields, demonstrating that geometry itself can control particle flow.

TitanA16 applies this principle at macroscopic scale: disordered conductive particles (e.g., copper granules) are constrained by a spiral or helical geometry, producing enhanced, reproducible conductivity without rare materials or superconductors.

Summary of the Invention

TitanA16 is a composite conductor in which chaotic conductive particles are guided into organized pathways by geometric constraints. The design:

Core: Random conductive particles (copper, copper alloys, or other metals)

Confinement layer: Spiral or helical geometry (wire, tape, or molded polymer)

Effect: Sharp, repeatable drop in resistance; hysteresis memory effect; scalable from small to large conductors

Advantage: Achieves controlled electron flow using only geometry; no exotic materials or external fields required

Detailed Description of the Invention

Material Preparation: Fill a transparent tube (glass or plastic) with random conductive granules.

Geometry Application: Wrap the tube in a tight helical or spiral layer to constrain particle alignment.

Electrical Connection: Connect a voltage source (e.g., 9V battery) to the ends of the tube.

Observation & Measurement: Conduct voltage mapping; observe hot and cold regions along the spiral indicating guided electron flow. Granules will naturally clump along the guided path, demonstrating geometry-driven alignment.

Optional Enhancements:

Outer layer or sheath to maintain structural integrity of spiral

Scaled designs for industrial or laboratory applications

Experimental Data: Demonstrates hysteresis, reproducible resistance drop, and improved conduction versus unconstrained granular media.

Claims

A composite conductor comprising disordered conductive particles constrained by a spiral or helical geometry to enhance electrical conductivity.

The conductor of claim 1, wherein the geometry produces a resistance flip threshold and a memory effect retaining enhanced conductivity.

The conductor of claims 1 or 2, wherein the conductive particles comprise copper, copper alloys, or other metals, scalable from small prototypes to industrial applications.

The conductor of any preceding claim, further comprising an outer layer or sheath to maintain geometric integrity and reduce drift.

Drawings / Figures

Fig. 1: Transparent tube with granular core and helical spiral confinement.

Fig. 2: Direction of current flow along spiral path; labeled: confinement layer, core particles, guided electron flow.

Fig. 3: Graph of resistance vs. time showing hysteresis behavior.

Next Steps

Have an attorney review and convert this skeleton into formal USPTO-compliant filing.

Include experimental PDFs, sketches, and validation data (Coherer analogy + Max Planck reference).

File as non-provisional utility patent under small entity fees if eligible.